

MICHAEL MASENHEIMER

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EDUCATION & HONORS

The University of Arizona, *Bachelor of Science*, Tucson, AZ Sep 2023 - May 2027

- **Major:** Computer Science; **Minor:** Information Science, Technology, and Arts; **GPA:** 3.62
- **Relevant coursework:** Software and Web Development, Data Structures and Algorithms, OOP design
- **Honors:** Dean's List with Distinction (2 semesters), Arizona Excellence Award, Outstanding Ambassador Award

SKILLS

Languages/frameworks: Java, Python, C, C++, JavaScript, Bash, Spring Boot, NodeJS, PostgreSQL, MongoDB, Keras

OS & DevOps technologies: Linux, Docker, Git, Jira, Maven, REST APIs, Vercel

EXPERIENCE

IBM, *Backend Software Engineering Intern*, Tucson, AZ May 2026 - Present

- Built an **agentic BIA ingestion pipeline** for IBM's Storage Defender product, combining LLM-based sanitization, multi-modal OCR, and RTO/RPO extraction mapped to governance profiles
- Engineered a **LangGraph ReAct agent** with an **A2A protocol server** for autonomous governance gap analysis, tuned across **18 LLM configs** using a **61+** document adversarial test corpus and LLM-as-judge scoring
- Instrumented the pipeline with a typed **15+** field metrics layer and a **3-tier async pytest suite**; contributed reusable scoring utilities to a shared IBM test library used across multiple AI agents

CERN (University of Arizona), *Embedded Software Researcher*, Tucson, AZ Apr 2025 - Present

- Research **quantized feedforward neural networks** for the forward calorimeter of the **world's largest particle accelerator**, to determine relevant proton collisions
- Assess custom network design and scope to benchmark latency, throughput, and quantization tradeoffs, achieving **~7 million packets/second throughput** and **97% accuracy**
- Develop and maintain kernel tile code and programmable logic on AMD AI engines to orchestrate tile data flow and **minimize bit-level overhead in C++**

University of Arizona Libraries, *Maker Studio Fabrication Lead*, Tucson, AZ Jan 2025 - Present

- **Lead a team of 8** studio assistants to run all 3D printing, laser cutting, CNC Milling and woodshop operations, managing weekly meetings and maintenance for **25** machines
- Assist **~25** students weekly with certifications in 3D printing and CNC machining; provide studio technical support for **40+ students weekly**

PROJECTS

CATalyst Studios Scheduling | Node, Express, React, MongoDB | [GitHub](#) May 2026 - Present

- Building a scheduling platform to **replace a legacy paid service**, saving **~\$350/year** while adding maker-studio-specific scheduling logic and full workplace Slack integration
- Developing drag-and-drop scheduling management, shift request workflows to be used by **~20** employees daily
- Secured the app with **JWT auth**, role-based access control, and **server-side rate limiting** across the stack

ECOMAP | Node, Express, Python, MongoDB | [\[HackArizona 2026\] GitHub](#) Apr 2026 - May 2026

- Architected a **full-stack, cloud-based** application connected to Google Maps API to optimize trips with up to **5** combined modes of transportation, with eco score leaderboards and a social-media style UI
- Developed backend with **MongoDB and Node.js**, implementing **20 REST API endpoints** for automated eco-point distribution, real-time leaderboards with **web sockets**, and session-based authentication logic

LEADERSHIP

Computer Science Lead Ambassador, Tucson, AZ Apr 2025 - Present

- **Co-Lead a team of 15** ambassadors, work on outreach and event scheduling, handle logistics for **~20** outreach/workshop events per semester
- **Lead department tours** and outreach events; volunteer at student career and development activities
- Develop interactive **OpenCV**-based tabling activities for **2000+** students and **10** local K-12 schools